

Maya Beach

mbeach@andrew.cmu.edu | linkedin.com/in/mayabeach | mayabeach.github.io/portfolio/

Education

Carnegie Mellon University, Pittsburgh, PA

Expected December 2026

- Bachelor of Science in Mechanical Engineering | Minor in Soft Technologies
- Overall GPA: 3.86/4.0
- Honors: Emerging Scholar in Mechanics and Design (2026)
- Coursework: Soft Robotics: Mechanics, Design and Modeling, Electromechanical Systems Design, Mechanical Systems Experimentation, Dynamic Systems and Control, Heat Transfer, Fluid Mechanics, Thermodynamics

Skills

Software: SolidWorks, Onshape, ANSYS Fluent and Mechanical, MATLAB, Python, Arduino, Microsoft Office, Google Drive

Technical: DFM, Manual Machining, Silicone Casting, GD&T, Tolerance Stackups, Root Cause Analysis, PFMEA, PLM/PDM

Experience

Symbolic

May 2026 - August 2026

Mechanical Engineering Intern

- Developed compliant silicone grass to prevent packages from rolling on a mobile robot. Created a stable CAD design that supported 300+ feature variations, designed and 3D printed molds, cast silicone prototypes, and implemented testing plan.
- Designed a die cast part to replace a machined component, resulting in an estimated cost reduction of ~\$15 per part (80%).
- Designed multiple iterations of adhesive gaskets to protect robot from payload spills and minimize the need for RTV.
- Traveled to customer site to perform mechanical and PCBA reworks on 60+ bots over 4 days.

Locus Robotics

May 2025 - August 2025

Product Quality Intern

- Supported in problem statement development and frequency characterizations across 90+ customer sites using robot error reporting and customer reported symptoms (Salesforce). Collated into 8D report for major product escalation.
- Worked cross-functionally with Product Management, Test Engineering, and Data Science to determine the effectiveness of proactive failure monitoring in the field and in-house refurbishing processes.
- Developed work instructions and technical documentation for internal rework processes and contract manufacturers.
- Tested and troubleshooted returned bots, PCBAS, LiDARs, and internal bot computers.
- Determined assembly risks by shadowing manufacturing. Generated a PFMEA report for manufacturing and quality engineers.

Social Decision Science, Carnegie Mellon University

September 2025 - December 2025

Teaching Assistant for Human Intelligence and Human Stupidity

- Designed tasks to assess human intelligence and group dynamics for the final exam.
- Created a custom controller using Raspberry Pi Pico that allows five people to play Mario Kart simultaneously.

AIPEX Lab, Carnegie Mellon University

February 2025 - May 2025

Mechanical Design Research Assistant

- Designed a self-contained, 5 DOF, tendon driven robotic thumb for a robotic hand and upper arm using SolidWorks.
- 3D printed prototypes and assembled models with off-the-shelf parts.

Projects

Autonomous Metal Scrap Collecting Robot

September 2026 - Present

- Designing a mobile robot that collects and drops off metal scrap using magnets, Raspberry Pi, and LiDAR. [Team of 5]

Accessible Keyboard and Mouse Interface

September 2025 - December 2025

- Designed an alternative keyboard and mouse system comprised of a gesture-detecting glove and customizable joystick for users with fine motor impairment under \$200. Enables web navigation through broad hand, arm, or foot movements. [Team of 6]
- Material selection, iterative development, and sewing of custom bend sensors using piezoresistive fabric.
- Construction of wearable glove and interfacing with electronic components

Collapsible Chair

March 2024 - May 2024

- Designed, performed FEA, and machined a collapsible camping chair using human centered design.
- Chair had a weight of 2.7lb, volume of 3.2L, < 30 second set up time, and simulated FOS of 4.8 for 200lb person.
- Awarded most marketable design out of 36 groups. [Team of 4]